11.

$2Al + Cr_2O_3 \rightarrow 2Cr + Al_2O_3$

$\Delta H_r^0 = [2\Delta H_f(Cr) + \Delta H_f(Al_2O_3)]$

$- [2\Delta H_f(Al) + \Delta H_f(Cr_2O_3)]$

$\Delta H_r^0 = [0 + (-1596\ kJ)]$

$- [0 + (-1134)]$

$\Delta H_r^0 = -1596\ kJ + 1134\ kJ$

$\Delta H_r^0 = -462\ kJ$

14.

$\Delta U = q + w$

$\Delta U = -1\ kJ + 4\ kJ$

$\Delta U = +3kJ$

15.

$Fe + 2HCl \rightarrow FeCl_2 + H_2$

ஒரு மோல் இரும்பு 1 மோல் ஹைட்ரஜன் வாயுவை வெளியேற்றுகிறது.

55.85 g இரும்பு = 1மோல் இரும்பு

$\therefore n = 1$

$T = 25^0\ C = 298\ K$

$w = -P\Delta V$

$$w = -P\left(\frac{nRT}{P}\right)$$

$w = -nRT$

$w = -1 \times 8.314 \times 298\ J$

$w = -2477.57\ J$

$w = -2.48\ kJ$

16.

$T_i = 125^0\ C = 398\ K$

$T_f = 25^0\ C = 298\ K$

$\Delta H = nC_p(T_f - T_i)$

$$\Delta H = 2 \times \frac{5}{2} R\ (298 - 398)$$

$\Delta H = -500\ R$

17.

$C + O_2 \rightarrow CO_2\ \Delta H^0 = -a\ KJ$ ..... (i)

$2CO + O_2 \rightarrow 2CO_2 \quad \Delta H^0 = -b\ kJ$ ..... (ii)

$$C + \frac{1}{2}O_2 \rightarrow CO\ \Delta H^0 = ?$$

(i) × 2

$2C + 2O_2 \rightarrow 2CO_2 \quad \Delta H^0 = -2a\ kJ$ ..... (iii)

சமன்பாடு (ii) ஐ திருப்பி எழுதுக

$2CO_2 \rightarrow 2CO + O_2 \quad \Delta H^0 = +b\ kJ$ ..... (iv)

(iii) + (iv)

$2C + O_2 \rightarrow 2CO\ \Delta H^0 = b - 2a\ kJ$ ..... (v)

(v) ÷ 2

$$C + \frac{1}{2}O_2 \rightarrow CO\ \Delta H^0 = \frac{(b-2a)}{2}\ kJ$$

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